Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form. Every question's wording is listed beside its number below, and the I do and We do model answers match the lesson page word for word.

Starter

1. Both a flat disc and a solid sphere look circular from the front (0°). Which test distinguishes them fairly?

Option A is right. Turn both objects to 45° and 90° at equal distance to compare their outlines.

I do model answer

1. How do we compare a disc and a sphere across viewpoints? Why do we describe Earth as approximately spherical?

Follow four steps in the Shape Evidence Lab. Step 1 (Front view at 0°): Both the flat disc and the sphere show a circular outline. One circular view cannot prove an object is a sphere. Step 2 (Oblique view at 45°): Keep viewing distance equal. At 45°, the disc looks oval while the sphere stays circular. Step 3 (Edge-on view at 90°): Turn both objects 90°. The disc shows a narrow edge while the sphere stays circular. Step 4 (Space evidence). Multi-viewpoint spacecraft photos show the Sun, Earth, and Moon are approximately spherical. Earth has mountains and a slight equatorial bulge. A classroom ball is a geometric model, not real planetary proof.

We do model answer and checkpoint

1. Inspect Cards A to D in the Shape Viewpoint Comparator. Match each card to its evidence role. Then repair the over-claimed statement.

Card A (Front view 0°: both look circular) shows why one view is inconclusive. Card B (45° and 90°: disc becomes oval then narrow) separates a disc from a sphere. Card C (Sphere in three views: circular at 0°, 45°, and 90°) models a 3D round body. Card D (Model limit and space evidence) explains why Earth is approximately spherical. Repair: do not claim one front photo proves a smooth sphere. Compare multiple viewpoints and state approximately spherical.

2. A flat disc and a solid ball turn 90° at equal distance. What outlines do we observe?

Option A: The flat disc looks narrow, while the solid ball keeps a circular outline.

Arrival · Supported

1. **1. Look at Box 1 on the left of the arrival drawing. Which object is a three-dimensional solid: the ball or the flat circle?** — The ball is a three-dimensional solid.
2. **2. Look at Box 1 again. If a mystery object looks circular from the front (0°), must it be a sphere?** — No. A flat disc also looks circular from the front.
3. **3. Look at Box 2 in the middle of the drawing. When the flat disc turns edge-on (90°), does its outline look narrow or wide?** — Its outline looks narrow when turned edge-on at 90°.
4. **4. Look at Box 3 on the right of the drawing. For a fair comparison, should both objects be held at the same distance?** — Yes. Keep both objects at the same viewing distance and angle.

Arrival · Independent

1. **1. Use Box 1 of the arrival drawing. Distinguish a three-dimensional sphere from a two-dimensional flat disc.** — A sphere has three-dimensional depth in all directions, while a flat disc is thin.
2. **2. Look at Box 1. Does one circular front view at 0° prove an object is a sphere?** — Both a flat disc and a solid sphere have a circular outline at 0°.
3. **3. Look at Box 2. How does a flat disc outline change at 45° and 90°?** — The disc becomes oval at 45° and narrow at 90°, whereas the sphere stays circular.
4. **4. Look at Box 3. Why do scientists call Earth 'approximately spherical' rather than a perfect smooth sphere?** — Earth has mountains and a small equatorial bulge, so it is slightly flattened.

Arrival · Aspire

1. **1. Use Box 1 of the arrival drawing. Explain geometrically why a flat disc and a sphere share the same outline at 0°.** — At 0°, the observer views the circular face perpendicular to the line of sight.
2. **2. Look at Box 2. Why does a 90° turn rule out a flat disc when an outline stays circular?** — A flat disc shows a narrow edge at 90°. A sphere stays circular from every angle.
3. **3. Look at Box 2 and Box 3. Name the two variables that must be controlled during a viewpoint comparison.** — Keep viewing distance and viewing angles (0°, 45°, 90°) identical for both objects.
4. **4. Look at Box 3. Distinguish what a classroom ball test shows from real spacecraft evidence of Earth.** — The ball models 3D geometry. Spacecraft images provide empirical evidence of the real Earth.

Arrival · Choice

1. **1. Look at Box 1 of the arrival drawing. Which object is three-dimensional: a ball or a flat card?** — A ball is three-dimensional.
2. **2. At 0° (front view), do both the disc and the ball look circular: Yes or No?** — Yes. Both look circular from the front.
3. **3. Look at Box 2. At 90° (edge-on), which object looks narrow: the flat disc or the ball?** — The flat disc looks narrow at 90°.
4. **4. Look at Box 3. Is Earth a perfect smooth ball or approximately spherical?** — Earth is approximately spherical.

You do · Supported mark scheme

1. **1. Use Reference A. Compare the flat disc and the solid sphere at 0° (front), 45° (oblique), and 90° (edge-on). Write or point to the outline for each view.** — At 0°, both look circular. At 45°, the disc looks oval and the sphere looks circular. At 90°, the disc looks narrow and the sphere looks circular. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference A. Complete: 'At 90°, the flat disc looks ___, while the sphere stays ___. We keep viewing distance the ___.'** — At 90°, the flat disc looks narrow, while the sphere stays circular. We keep viewing distance the same. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. Use Reference B. Explain why we say Earth, the Moon, and the Sun are 'approximately spherical.' State one limit of the classroom ball.** — They are nearly round, but Earth has mountains and a slight bulge. Limit: a classroom ball is a model, not real space evidence. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Independent mark scheme

1. **1. Use Reference A. Record the outlines of a flat disc and a sphere across three controlled viewpoints (0°, 45°, and 90°). State the two fairness controls.** — At 0° both are circular; at 45° the disc is oval and the sphere is circular; at 90° the disc is narrow and the sphere is circular. Controls: same distance and same angles. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference A. Why does one circular front photo fail to prove a space body is a sphere?** — A flat disc and a sphere both produce a circular outline from the front (0°), so multiple viewpoints are required to distinguish them. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B. Link multi-viewpoint space evidence and one model limit. Explain why Earth is called approximately spherical.** — Spacecraft observations from multiple viewpoints show Earth is round, and 'approximately spherical' accounts for mountains and the equatorial bulge. Limit: a desk ball omits real scale. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Aspire mark scheme

1. **1. Use Reference A. Contrast the outlines of a flat disc and a sphere at 0°, 45°, and 90°. Keep viewing distance equal.** — Because a disc lacks depth, rotating it from 0° to 45° and 90° foreshortens its circular face into an ellipse and then a narrow strip. A sphere's symmetry keeps a circular outline at every viewpoint. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference B. Evaluate the claim: 'One circular image proves a body is a perfect sphere.' Separate observation from interpretation.** — Observation: one image shows a circular 2D outline. Interpretation error: a flat disc at 0° shares that outline, and real planets are approximately spherical (oblate) with surface relief. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B. Link model evidence and spacecraft views to justify calling Sun, Earth, and Moon approximately spherical. State one model limit.** — The ball model predicts constant circular outlines across angles, matching multi-viewpoint spacecraft imagery of the Sun, Earth, and Moon. 'Approximately spherical' reflects Earth's equatorial bulge; the desk ball lacks real scale and gravity. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Choice mark scheme

1. **1. 1. Look at Reference A. At 0° (front view), do the disc and sphere look the same (circular) or different? 2. At 90° (edge-on), which object looks narrow: the disc or the sphere?** — 1. They both look circular at 0°. 2. The disc looks narrow at 90°. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. 3. Should viewing distance be the same or different for a fair test? 4. Can one front photo alone prove a sphere: No or Yes?** — 3. Viewing distance must be the same. 4. No,

because a disc also looks circular from the front. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. 5. Look at Reference B. Which phrase best describes Earth, Sun, and Moon: approximately spherical or flat discs? 6. Name one limit of a smooth ball model: omits mountains/scale, or is real Earth?** — 5. Approximately spherical. 6. It omits mountains and real planetary scale. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

Exit ticket · Supported

1. **1. When turned edge-on at 90°, which object becomes narrow: a flat disc or a sphere?** — A flat disc becomes narrow at 90°, while a sphere stays circular.
2. **2. Why do we say Earth is 'approximately spherical' instead of a perfect smooth ball?** — Earth has surface mountains and is slightly flattened with a small bulge.
3. **3. Is a classroom ball a helpful model of Earth or a real photograph from space?** — It is a helpful classroom model, not a real space photograph.

Exit ticket · Independent

1. **1. Describe one observation across viewpoints (0°, 45°, 90°) that separates a flat disc from a sphere.** — When turned from 0° to 90°, a flat disc changes from circular to narrow, whereas a sphere keeps a circular outline.
2. **2. Why must viewing distance and angle stay equal when comparing the disc and sphere?** — Keeping distance and angle equal ensures a fair comparison so only the object's shape causes any outline change.
3. **3. Why are the Sun, Earth, and Moon called approximately spherical? State one model limit.** — Space observations show they are nearly round with slight flattening and surface features; a smooth desk ball omits scale and relief.

Exit ticket · Aspire

1. **1. Why are multi-viewpoint spacecraft photos stronger evidence than one circular photo?** — One circular view fits a flat disc at 0°, whereas circular outlines from multiple viewpoints confirm three-dimensional spherical geometry.
2. **2. Explain why describing Earth as 'approximately spherical' is more accurate than 'a perfect sphere'.** — Earth's rotation causes a small equatorial bulge (oblate spheroid) and its crust has mountains and valleys.
3. **3. State what the classroom disc-and-sphere comparison proves and what it cannot prove alone.** — It proves how 2D and 3D shapes project across angles, but real planetary shape requires empirical space observations.

Exit ticket · Choice

1. **1. At 90° (edge-on), does a sphere stay circular or turn into a thin line?** — A sphere stays circular at 90°.
2. **2. Does one front circle prove a sphere: No (a disc is circular too) or Yes?** — No. A flat disc also looks circular from the front.
3. **3. Are the Sun, Earth, and Moon flat discs or approximately spherical bodies?** — They are approximately spherical bodies.

Exit pupil voice

1. What helped you today, and what would you change? — Accept any pointed, spoken, signed, drawn, written, or chosen reflection on what helped and what could change next time. A pupil may pass.

Topic guidance and Key Stage 3 progression

- **Three-viewpoint geometry at Key Stage 3:** At **0° (front)** both a flat disc and a sphere show a circular outline. At **45° (oblique)** the disc foreshortens into an oval, and at **90° (edge-on quarter-turn)** the disc becomes narrow while the sphere stays circular.
- **Controlled comparison:** Both objects must be observed at the same distance and the same angles so only three-dimensional shape causes the outline difference.
- **Approximately spherical & model limits:** Multi-viewpoint spacecraft observations show the Sun, Earth, and Moon are **approximately spherical** (Earth is an oblate spheroid with mountains and an equatorial bulge; a classroom ball models 3D geometry but is not itself proof of Earth's shape).

Made by Matt